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"""
ARKHEIA-CPS – Explanation Generator
Produces plain-language, consumer-first
explanations for every violation type.
Tone: protective, empowering, non-
technical.
"""

from __future__ import annotations
from decimal import Decimal
from typing import List, Optional

from app.schemas.common import Alert

# — AUTO explanations
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def generate_auto_explanations(
    alerts: List[Alert],
    perfection_days: Optional[int],
    illegal_patterns: List[str],
) -> List[str]:
    explanations = []

    # Perfection timing narrative
    if perfection_days is not None:
        if perfection_days > 90:
            explanations.append(
                f"The lender perfected the

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lien {perfection_days} days after purchase.
"

        f"Typical perfection occurs
within 10-30 days. This severe delay may
affect "

        f"the legality of any
repossession or enforcement action taken
against you. "

        f"Consult an attorney
before making any payments or surrendering
the vehicle."
    )
    elif perfection_days > 60:
        explanations.append(
            f"The lender perfected the
lien {perfection_days} days after purchase
- "

            f"a high deviation from the
expected 10-30 day window. This may create
"

            f"questions about the
enforceability of the lender's security
interest."
        )
    elif perfection_days > 30:
        explanations.append(
            f"The lien was perfected
{perfection_days} days after purchase. "
            f"This is slightly above
the typical 10-30 day window. "
            f"Request documentation
confirming the perfection date from your
lender."
        )
    )
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# Illegal pattern narratives
for pattern in illegal_patterns:
    if "PERF_AFTER_REPO" in pattern:
        explanations.append(
            "CRITICAL: The lender
repossessed your vehicle before their lien
was legally "
            "perfected. This means they
had no legal right to take the vehicle. "
            "You may be entitled to the
return of the vehicle and damages. "
            "Contact a consumer
protection attorney immediately."
        )
    if "REPO_NO_NOTICE" in pattern:
        explanations.append(
            "CRITICAL: Your vehicle was
repossessed before you received a notice of
"
            "default. In most states,
lenders are required to notify you before
taking "
            "action. This repossession
may be unlawful. Do not pay any deficiency
"
            "balance until you speak
with an attorney."
        )
    if "REPO_BEFORE_CURE" in pattern:
        explanations.append(
            "CRITICAL: The lender did
not give you enough time to cure the
default "
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        "before repossessing the
vehicle. Georgia law requires lenders to
give you "
            "a right-to-cure notice and
adequate time to catch up on payments. "
            "This repossession may
violate O.C.G.A. § 10-1-36."
        )
        if "REPO_CURE_TOO_SOON" in pattern:
            explanations.append(
                "The lender did not wait
the legally required number of days after
sending "
                "the right-to-cure notice
before repossessing the vehicle. "
                "The cure period was not
fully honored."
            )
        if "PERF_SAME_DAY" in pattern:
            explanations.append(
                "The lender perfected the
lien and repossessed the vehicle on the
same day. "
                "This is a predatory
pattern that suggests the repossession was
planned "
                "before the lien was
legally established. Seek legal counsel."
            )

    # Alert-based narratives (non-
perfection)
    for alert in alerts:
        code = alert.code

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        if "AFFORDABILITY" in code or
        "PAYMENT_TO_INCOME" in code:
            explanations.append(
                "Your monthly payment
appears to exceed what is typically
considered affordable "
                "based on your income.
Financial experts recommend that car
payments stay "
                "below 15% of your take-
home pay. At this level, you may have
difficulty "
                "covering other essential
expenses."
            )
        elif "APR" in code and "PREDATORY"
not in code:
            explanations.append(
                f"The interest rate on this
loan is above the typical range for someone
with "
                f"your credit profile. A
high APR means you will pay significantly
more "
                f"over the life of the
loan. Shop around at credit unions and
banks – "
                f"you may qualify for a
better rate."
            )
        elif "UNDISCLOSED" in code:
            explanations.append(
                "One or more fees on this
contract were not clearly disclosed to you
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before "
            "signing. Georgia law and
            federal Truth-in-Lending rules require that
            all "
            "fees be clearly itemized.
            Request a full fee disclosure in writing."
        )
        elif "FORCED" in code or "ADDON" in
code:
            explanations.append(
                "One or more add-on
products (such as extended warranties or
GAP insurance) "
                "appear to have been
presented as required. These products are
always optional "
                "under Georgia law. You
have the right to decline them."
            )
        elif "TOTAL_COST" in code:
            explanations.append(
                "The total cost stated in
the contract does not match what the
payment "
                "schedule would add up to.
This discrepancy may indicate hidden
charges "
                "or a math error in your
contract. Request a detailed payoff
schedule."
            )

        # Deduplicate
        seen = set()

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        unique = []
        for e in explanations:
            key = e[:60]
            if key not in seen:
                seen.add(key)
                unique.append(e)

        return unique

# — HOUSING explanations
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def generate_housing_explanations(
    alerts: List[Alert],
    payload,
    illegal_clauses: List[str],
    total_movein: Decimal,
) -> List[str]:
    explanations = []
    rent = payload.base_monthly_rent

    # Move-in cost narrative
    months_equiv = float(total_movein /
rent) if rent > 0 else 0
    if months_equiv > 3.0:
        explanations.append(
            f"Your total move-in cost is
${total_movein:,.2f} – that's
{months_equiv:.1f} "
            f"months of rent upfront.
Amounts above 3× monthly rent are
considered predatory "

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        f"in most jurisdictions and may
violate state security deposit limits."
    )
    elif months_equiv > 2.5:
        explanations.append(
            f"Your total move-in cost of
${total_movein:,.2f} ({months_equiv:.1f}x
rent) "
            f"is above the typical 2x
guideline. Ask the landlord to itemize
every charge."
        )

    # Illegal clause narratives
    if illegal_clauses:
        explanations.append(
            f"This lease contains
{len(illegal_clauses)} clause(s) that may
be illegal "
            f"or unenforceable under tenant
protection law: "
            f"{'; '.join(f'\"{c}\"' for c
in illegal_clauses[:3])}. "
            f"Clauses that waive your legal
rights, allow entry without notice, or
disclaim "
            f"habitability obligations may
not be enforceable even if you signed the
lease."
        )

    # Alert-based narratives
    for alert in alerts:
        code = alert.code

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        if "RENT_TO_INCOME" in code:
            ratio_pct = float(rent /
(payload.consumer.net_monthly_income or
payload.consumer.reported_monthly_income or
1)) * 100

            explanations.append(
                f"Your rent of
${rent:,.2f}/month is approximately
{ratio_pct:.0f}% of your "
                f"take-home income. Housing
experts recommend spending no more than 30%
of "
                f"net income on rent. At
this level, you may face financial
hardship."
            )

        elif "APPLICATION_FEE" in code:
            explanations.append(
                f"The application fee of
${payload.application_fee_amount:,.2f}
exceeds "
                f"typical limits ($25-$75).
High application fees can be a predatory
tactic, "
                f"especially if the
landlord collects fees from many applicants
with no "
                f"intention of renting to
all of them."
            )

        elif "ADMIN_FEE" in code:
            explanations.append(
                f"The administrative fee of
${payload.admin_fee_amount:,.2f} is above "

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        f"the typical range ($0-
$150). Ask the landlord to explain exactly
what "

        f"service this fee covers."
    )
    elif "SECURITY_DEPOSIT" in code:
        explanations.append(
            f"The security deposit of
${payload.security_deposit_amount:,.2f} may
"

            f"exceed the legal limit in
your state. Many states cap security
deposits "

            f"at 1-2 months' rent.
Check your state's tenant protection laws."
        )
    elif "RENT_INCREASE" in code:
        explanations.append(
            "The rent increase clause
in this lease allows for above-average rent
hikes. "

            "High rent increase caps
can make your housing costs unpredictable
over time. "

            "If you sign a multi-year
lease, negotiate a lower cap in writing."
        )
    elif "LATE_FEE" in code:
        explanations.append(
            f"The late fee of
${payload.late_fee_amount:,.2f} is above
the recommended "

            f"maximum. Excessive late
fees can quickly compound if you miss a

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payment, "
            f"making it harder to catch
up."
        )
        elif "GRACE_PERIOD" in code:
            explanations.append(
                "This lease has no grace
period or a very short one before late fees
apply. "
                "A grace period of at least
3-5 days protects you if rent processing is
delayed."
            )
        elif "AUTO_RENEWAL" in code:
            explanations.append(
                "This lease automatically
renews without adequate advance notice. "
                "If you don't plan to
renew, you could be locked into another
term. "
                "Mark your calendar at
least 60 days before the lease ends."
            )
        elif "EARLY_TERMINATION" in code:
            explanations.append(
                "The early termination
penalty in this lease is high. If you need
to "
                "leave before the lease
ends for any reason, you could owe several
months' "
                "rent. Negotiate this
clause before signing."
            )

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        elif "EVICTION" in code:
            explanations.append(
                "The eviction clause in
this lease may allow the landlord to begin
eviction "
                "proceedings faster than
state law normally permits. Know your
rights: "
                "most states require proper
notice and a court hearing before you can
be "
                "legally removed."
            )
        elif "ILLEGAL_CLAUSE" in code:
            explanations.append(
                "One or more clauses in
this lease may be illegal and
unenforceable. "
                "Signing a lease does not
mean you waive rights that are protected by
law. "
                "Consult a tenant rights
organization or attorney before signing."
            )

# Deduplicate
seen = set()
unique = []
for e in explanations:
    key = e[:60]
    if key not in seen:
        seen.add(key)
        unique.append(e)

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return unique
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